IMPERIAL COLLEGE LONDON

DEPARTMENT OF ELECTRICAL AND ELECTRONIC ENGINEERING
EXAMINATIONS 2019

EEE/EIE PART I: MEng, BEng and ACGI

INTRODUCTION TO SIGNALS AND COMMUNICATIONS

Corrected Copy

Wednesday, 29 May 10:00 am

Time allowed: 2:00 hours

There are THREE questions on this paper.

Answer ALL questions.
Q1 carries 40% of the marks. Questions 2 and 3 carry equal marks (30% each).

Any special instructions for invigilators and information for candidates are on page 1.

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Special Instructions for Invigilator: **None**

Information for Students:

Some useful trigonometric identities

$$\cos x \cos y = \tfrac{1}{2}\cos(x-y) + \tfrac{1}{2}\cos(x+y)$$
$$\sin(x-y) = \sin x \cos y - \sin y \cos x$$
$$a\cos x + b\sin x = c\cos(x+\theta)$$

where $c = \sqrt{a^2 + b^2}$ and $\theta = \tan^{-1}(-b/a)$

Complex exponential

$$e^{jx} = \cos x + j \sin x$$

1. This is a general question. (40%)

a. Fourier transforms of signals.

i. Given the Fourier transform $F_1(\omega) = 2\pi\delta(\omega - \omega_o)$, what is the corresponding time-domain signal $f_1(t)$? [2]

ii. What is the time-domain signal $f_2(t)$ corresponding to the Fourier transform of $F_2(\omega) = 2\pi\delta(\omega + \omega_o)$? [2]

iii. Using results in parts i and ii, find the Fourier transform of $g(t) = \cos(\omega_o t)$. [2]

iv. From result in part iii, what can be said about the frequency spectrum of $g(t)$? [1]

b. A periodic signal $g(t)$ with period T is given by the following Fourier series:

$$g(t) = a_0 + \sum_{n=1}^{\infty} a_n \cos(n\omega_0 t) + b_n \sin(n\omega_0 t).$$

i. Show that $\cos(m\omega_o t)$ and $\sin(n\omega_o t)$ are orthogonal to each other for all m and n with $m \neq n$. [4]

ii. Derive an expression for the power of $g(t)$ in terms of a_0, a_n and b_n for $n = 1, 2, ..., \infty$ and explain how result in part i is used in obtaining the expression. [3]

iii. Why can $g(t)$ be represented by the aforementioned Fourier series? [2]

iv. If $g(t)$ is an odd function of t, what can be said about any of the coefficients, a_0, a_n and b_n for $n = 1, 2, ..., \infty$, and why? [3]

c. Let $m(t)$ and $M(\omega)$ denote a real-valued, energy signal with bandwidth of B Hz and its Fourier transform, respectively. The signal $m(t)$ is used to modulate a sinusoidal carrier $\cos(\omega_c t)$ to form the full amplitude modulation (AM) signal $\phi(t)$ where $\omega_c >> 2\pi B$ and the amplitude of the carrier signal included in $\phi(t)$ is A.

i. Give an expression for the full AM signal $\phi(t)$. [2]

ii. Sketch the spectrum of $\phi(t)$ obtained in part i. [2]

iii. What is the bandwidth of $\phi(t)$ in Hz? [1]

iv. Give a condition in terms of A and $m(t)$ that ensures the applicability of envelope detection at the receiver, and explain why. [2]

v. Let E_m denote the energy of $m(t)$. If the carrier signal is removed from $\phi(t)$, what is the energy of the remaining part of $\phi(t)$ and why? [3]

d. Let $\varphi(t)$ denote the frequency-modulation (FM) waveform of the modulating signal $m(t)$ where k_f is used to denote the proportionality constant for the frequency deviation, and A and ω_C are the amplitude and angular frequency of the carrier (in

radians per second), respectively. The bandwidth of $m(t)$ is B Hz. To ease our notation, let us define $a(t) = \int_0^t m(u)\,du$.

i. Give an expression for the FM waveform $\varphi(t)$. [2]

ii. Give an expression for the complex FM waveform $\hat{\varphi}(t)$ such that $\varphi(t) = \mathrm{Re}[\hat{\varphi}(t)]$. [3]

iii. By using the Taylor expansion of $\hat{\varphi}(t)$ from part ii and considering a condition $|k_f a(t)| << 1$, derive an expression to approximate $\varphi(t)$. [3]

iv. What is the physical meaning of the condition $|k_f a(t)| << 1$? [1]

v. From result in part iii, sketch the spectrum and identify the bandwidth of the approximate $\varphi(t)$ under the condition of $|k_f a(t)| << 1$. Justify your result. [2]

2. Signal convolution, spectral power density and autocorrelation. (30%)

a. Consider the following signal $x(t)$ where T is a positive constant:

$$x(t)=\begin{cases} 0 & t<0 \\ 1 & 0\le t\le T \\ 0 & t>T \end{cases}$$

Let $y(t)$ be the self-convolution of $x(t)$. That is, $y(t)=x(t)*x(t)$.

i. Express $y(t)$ as an integral of $x(t)$. [2]

ii. Carry out the convolution integration to obtain and sketch $y(t)$. [8]

iii. Let $X(\omega)$ and $Y(\omega)$ denote the Fourier transforms of $x(t)$ and $y(t)$, respectively. How are $X(\omega)$ and $Y(\omega)$ related to each other? Prove their relationship. [6]

b. Consider a real-valued, energy signal $g(t)$ in time domain and its Fourier transform $G(\omega)$. Let $\psi_g(\tau)$ denote the autocorrelation function of $g(t)$ where τ is the time lag.

i. Let E_g denote the energy of $g(t)$. Give an expression for E_g in terms of $g(t)$. [2]

ii. Using the result in part i, show that $E_g=\frac{1}{2\pi}\int_{-\infty}^{\infty}|G(\omega)|^2 d\omega$. [4]

iii. Express $\psi_g(\tau)$ as a function of $g(t)$ and τ. [2]

iv. Show that the Fourier transform of $\psi_g(\tau)$ is equal to $|G(\omega)|^2$. [4]

v. What is the physical meaning of $\psi_g(\tau)$? Since $g(t)$ is not periodic (as it is an energy signal), what is the typical value of $\psi_g(\tau)$ when $\tau\to\infty$? [2]

3. Communications techniques. (30%)

a. A signal $g(t)$ with bandwidth of BHz is sampled periodically at a frequency of f_S Hz to obtain the samples $\tilde{g}(t)$.

i. What is the minimum value of f_S such that $g(t)$ can be properly recovered from the samples $\tilde{g}(t)$? Draw a spectrum diagram to explain. [4]

ii. Suppose that a steady stream of signal samples at a rate of R samples per second is received. Using the result in part i, what is the maximum possible bandwidth of the signal represented by the samples? Explain why. [3]

iii. Using the result in part ii, what is the maximum rate of samples (i.e., the maximum number of samples per second) that a noiseless communication channel of L Hz can transmit without loss of information? Explain why. [3]

b. Design a communications system to transmit a TV signal consisting of a picture and sound signals, $m_p(t)$ and $m_s(t)$, which have a bandwidth of 5 MHz and 0.5 MHz, respectively. The system transmits the TV signal over the frequency band from 44 to 56 MHz, where the picture and sound signals occupy two contiguous and non-overlapping parts of the spectrum. The transmitted TV signal is denoted by $\phi(t)$. Two sinusoidal (carrier) signals of frequencies at 5.5 and 50 MHz, which are synchronized in phase and frequency, are available at the transmitter and receiver.

i. Describe a method of amplitude modulation (AM) with double-side-band transmission by using the given sinusoidal signals to generate the TV signal for transmission in the 44-56 MHz frequency band. Sketch the frequency spectrum of the transmitted signal $\phi(t)$ and identify the spectrum components associated with the picture and sound signals. [6]

ii. Give an expression for the transmitted TV signal $\phi(t)$. [4]

iii. Assuming that ideal filters are available, draw a block diagram of the receiver and show mathematically how the picture and sound signals, $m_p(t)$ and $m_s(t)$, can be recovered by using the sinusoidal signals of 5.5 and 50 MHz at the receiver. [6]

iv. How should the system design be changed if the frequency band only from 50 to 56 MHz is available for the TV signal transmission? [4]